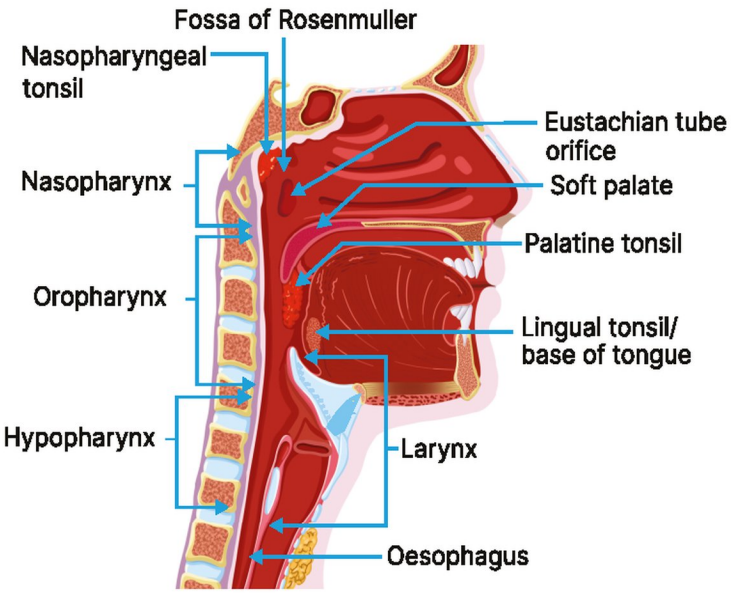
15. PHARYNX PART-1

ANATOMY OF PHARYNX

00:01:50

- Pharynx is a fibromuscular tube
- Extending superiorly from the base of the skull
- Inferiorly extends up to the lower border of the Cricoid cartilage corresponding to C6 vertebrae
- 3 subsegments of the pharynx: Nasopharynx, Oropharynx, & Hypopharynx



Nasopharynx

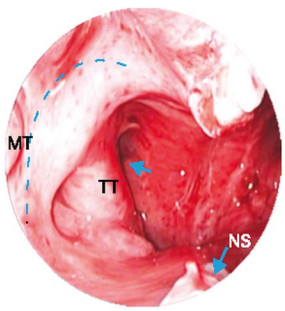
- Extends superiorly from the base of the skull and inferiorly up to the imaginary line that goes through the palate
- Corresponding vertebrae → C1

Boundaries

- Anteriorly → Communicates with the nasal cavity through choanae
- Posteriorly → Related to the C1 vertebrae
- Roof and posterior wall → Continuous with each other

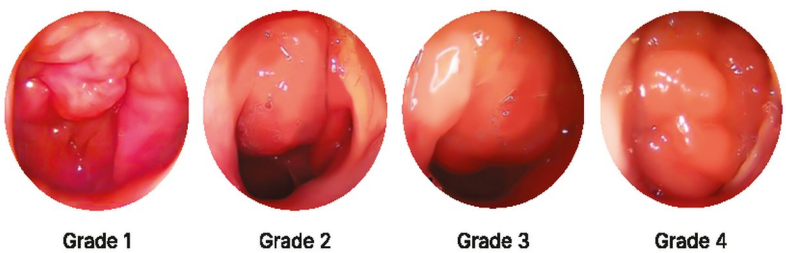
Structures Of The Lateral Wall

- Opening of the Eustachian tube (yellow arrow)
- Torus tubarius (TT)
- Fossa of Rosenmuller (blue arrow) → MC site of origin for Nasopharyngeal cancer
- Sphenopalatine foramen → Communication between the nasopharynx and pterygopalatine fossa



Adenoids

- Lymphoid tissue in the midline of the medial wall of the nasopharynx
- Especially seen in children
- Can be identified by endoscopically /X-ray /Portable endoscopes



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- Grading of adenoids- Based on the choanal space occupied
  - Grade 1- 0-25%
  - Grade 2- 25-50%
  - Grade 3- 50-75%
  - Grade 4- >75%

Passavant's Ridge

- Present between the nasopharynx and oropharynx
- Formed by the fibers of the palatopharyngeus muscle
- Prevents nasal regurgitation
- Injury during adenoidectomy/tonsilectomy leads to nasal regurgitation

Oropharynx

- Extends
  - Superiorly from the imaginary line that goes through the palate
  - Inferiorly up to the imaginary line passing through the tip of the epiglottis corresponding to the Hyoid bone
- Corresponding vertebrae → C2 & some portion of C3

Boundaries

- Anteriorly → Oral cavity
- Posteriorly → C2 vertebrae

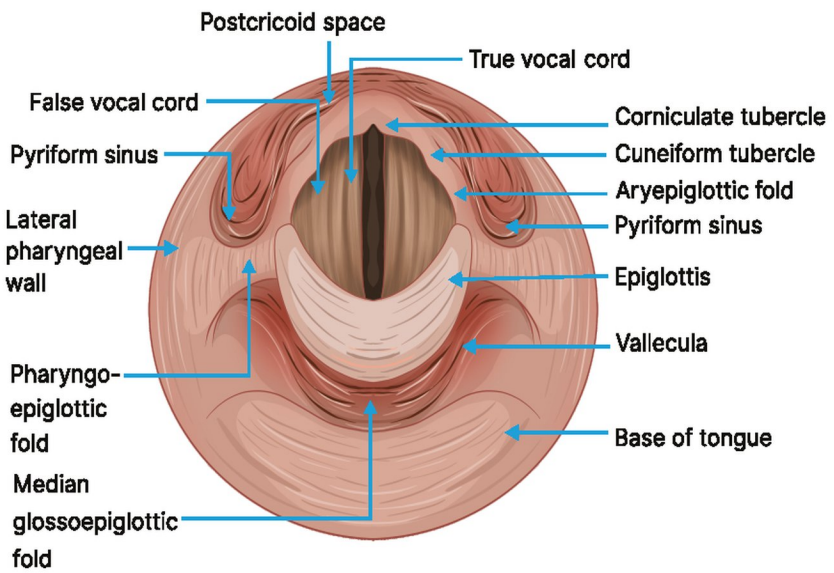
Structures of The Lateral Wall

- 2 pillars
  - Anterior → Formed by Palatoglossus
  - Posterio → Formed by Palatopharyngeus
- Palatine/Paucial/Pharyngeal tonsil → Lymphoid tissue between Palatoglossus & Palatopharyngeus



Contents

- Tonsil
- Posterior 1/3rd of the tongue (base)
- Median glossoepiglottic fold
- Lateral glossoepiglottic fold
- Valecula → Space between the median and lateral glossoepiglottic fold



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Oral Cavity

00:16:05

FMGE 2021
NEET PG 2018

- Boundaries of the oral cavity
  - Anteriorly → Lips
  - Posteriorly → Anterior pillar
  - Superiorly → Hard palate
  - Inferiorly → Anterior 2/3rd of the tongue (presses on the floor of the mouth)

Tongue Development And Nerve Supply

00:16:36

Anterior 2/3rd

- Derived from the 1st pharyngeal arch
- Formed by the fusion of:
  - Two lingual swellings
  - Tuberculum impar
- Nerve Supply (1st Arch)
  - General sensation → Lingual nerve (Branch of mandibular division of Trigeminal nerve)
  - Taste (special sensation) → Chorda tympani (Branch of Facial nerve)
- Pathology in Anterior 2/3rd (ulcer, cancer) → Referred otalgia
  - Mediated via Lingual nerve (branch of Trigeminal nerve)
  - The trigeminal nerve supplies the ear via the auriculotemporal nerve

Posterior 1/3rd

- Develops from the Hypobranchial eminence, which has two parts (Cranial and Caudal)
- Cranial Part
  - Derived from the 3rd arch
  - Forms most of the posterior 1/3rd (base of tongue)
  - General sensation → Glossopharyngeal nerve
- Caudal Part
  - Derived from the 4th arch
  - Forms the vallecula
  - General sensation → Vagus nerve

Motor nerve supply of the tongue

- All muscles of the tongue are supplied by the hypoglossal nerve
- Exception → Palatoglossus

Hypopharynx

00:21:50

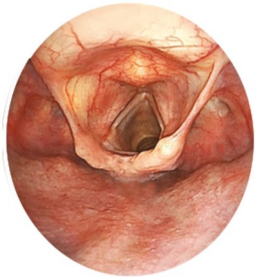
- Also called the laryngopharynx
- Lies behind the larynx

Boundaries

- Superiorly from an imaginary line through the tip of the epiglottis
- Inferiorly up to the lower border of the cricoid cartilage corresponding to C6 vertebrae
- Anterior → Related to larynx
- Posterior → Related to the vertebrae C3, C4, C5, and C6

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Subdivisions of the Hypopharynx



- Pyriform fossa
- Post cricoid region
- Post pharyngeal wall

Pyriform Fossa

- Superior laryngeal nerve (internal branch) traverses through the pyriform fossa
- Pathology of ILN
- E.g., A pt complains of impaction of a fish bone in the pyriform fossa. During the extraction, the SLN could be injured.
- ILN is sensory to the larynx above the level of the vocal cord → Injury leads to loss of sensation
- AKA, Smuggler's pouch

Post Cricoid Region

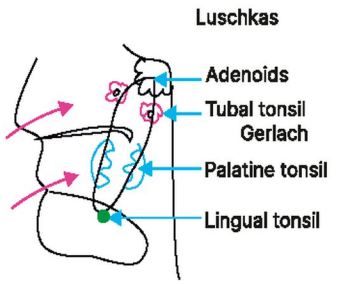
- MC site for carcinoma in Plummer-Vinson syndrome

Waldeyers Ring

00:27:03

- Ring of lymphoid tissue in the upper aero-digestive tract
- Protective tissue → Traps pathogens entering the body

Composition of Waldeyers Ring



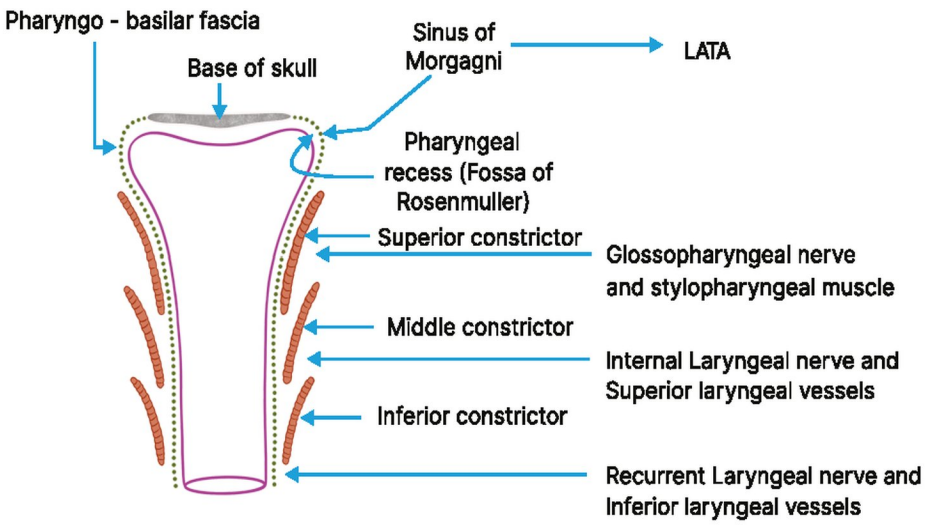
- Superiorly → Adenoids (AKA Luschka's tonsil)
- Lateral wall → Tubal tonsil (lymphoid tissue around the Eustachian tube, AKA Gerlach tonsil), Palatine tonsil
- Base of tongue → Lingual tonsil

Pharyngeal Wall Layers

00:28:56

- Innermost layer (1st layer) → Mucosa
- 2nd layer → Pharygobasilar fascia
- 3rd layer → Muscular layer; External row → Superior, middle, and inferior constrictor
- 4th layer → Buccopharyngeal fascia

Structures Passing Through The Pharyngeal Wall



Area	Structures passing through
Sinus of Morgagni • Gap between the base of the skull and the superior constrictor muscle • At the level of the nasopharynx	**(Mnemonic: LATA)** • Levator veli palatini (helps open ET) • Auditory tube (Eustachian tube) • Tensor veli palatini (helps open ET) • Ascending palatine artery
Between the superior and middle constrictor muscle • At the level of the oropharynx	• Glossopharyngeal nerve • Stylopharyngeal muscle
Between the middle and the inferior constrictor muscle • At the level of the hypopharynx	• Internal laryngeal nerve • Superior laryngeal vessels
Below the inferior constrictor muscle • At the level of the larynx	• Recurrent laryngeal nerve • Inferior laryngeal vessels

One Liners

00:35:10

- Lower limit of nasopharynx is imaginary line passing through the Palate
- Lower limit of oropharynx is imaginary line passing through the tip of the epiglottis corresponds to Hyoid bone
- Lower limit of hypopharynx is lower border of cricoid cartilage corresponding to C6 vertebra
- Capsule of the tonsil is formed by Phargobasilar fascia
- Killian's dehiscence is seen Between the fibers of inferior constrictor muscle (Oblique fibers of thyropharyngeus and horizontal fibers of cricopharyngeus)
  - Zenker's diverticulum is outpouching of Killian's dehiscence
- Passavant's ridge is formed by Palatopharyngeus
- Sinus of Morgagni lies between the base of the skull and the superior constrictor muscle
- Luschka's tonsil is the adenoids
- Gerlach tonsil is the tubal tonsil
- Eustachian tube opens in the nasopharynx approximately 1 cm behind the inferior turbinate
- Sphenopalatine foramen passes 8-10 mm behind the middle turbinate
- Lining epithelium of the adenoid is Pseudo-stratified ciliated columnar epithelium
- Lining epithelium of tonsil → Stratified squamous epithelium
- Crypts/Capsules are seen in the tonsil (Hemi-encapsulated)
- Bag of worm feel on palpation of Adenoid
  - The internal carotid artery (ICA) lies close to the adenoids
  - Palpation helps detect pulsations of an aberrant ICA
  - If strong pulsations are felt:
    → Surgery is stopped
    → Angiography is performed to identify the ICA position
- Hot potato voice is seen in Quinsy (peritonsillar abscess)
- Referred otalgia from
  - Oral lesions → From the 5th nerve
  - Oropharyngeal lesions → From the 9th nerve
  - Hypopharyngeal and laryngeal lesions → From the 10th nerve

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ADENOID HYPERTROPHY

00:41:45

FMGE 2020, 2021

- Age of presentation – 19-14 yrs

Presentation

- Adenoid facies → Pinched nose, a high-arched palate, crowded teeth, dull expression
- Nasal obstruction
- Sinusitis
- Middle ear disease (d/t obstruction of ET by adenoid hypertrophy)
- Sleep apnea (d/t adenoid & tonsillar hypertrophy)
- Rhinolalia clausa (nasal twang)

Diagnosis

- Endoscopy
- X-ray, lateral view of nasopharynx
  - Draw two imaginary lines:
    → One along the base of the skull
    → One along the hard palate
  - This divides the nasopharyngeal airway into 4 equal compartments (≈25% each)
    → Grade 1 → 0-25%
    → Grade 2 → 25-50%
    → Grade 3 → 51-75%
    → Grade 4 → >75%



Treatment

- Grade 1 & 2 → Conservative treatment with intranasal corticosteroids and antihistamines
- Grade 3 & 4 → Surgery
- Adenoid hypertrophy + Serous otitis media → Adenoidectomy + Myringotomy with Grommet

Indication For Surgery

- Absolute indications
  - Obstructive sleep apnea
  - Grade 3 & 4 hypertrophy
- Relative indications
  - Recurrent infections
  - Recurrent cold
  - Acute otitis media

Contraindication For Surgery

- Age <3 yrs (compromises the immunity)
- Cleft palate (causes lifetime nasal regurgitation)
- HB <10 mg/dl
- Active URTI
- Bleeding disorder

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ADENOID HYPOTROPHY

History and Examination

- Child with nasal obst., pinched nose, absent nasolabial crease, open mouth (PYQ), crowding of anterior upper teeth, high arched palate, (adenoid facles PYD sleep apnea.
- Recurrent infections (sinusitis, ADM)
- B/L decreased hearing due to SOM
- Rhinolalia clausa
- Failure to thrive (PYO

Investigations

- Endoscopy to assess the size.
- X-ray nose and nasopharynx lateral view IDodd/
  crescent sign-ve(PYQ)-crescent of air not present between mass and pharyngeal wall as he mass arises from the pharyngeal wall itself).
- B type tympanogram(PYQ)

Aden old hypertrophy-Important points to remember

Initial management
Medical

Surgical Management

- Adenoidectomy grommet insertion(PYQ)
- Position-Rose's
- Cold-Blind curettage, microdebrider
- Hot-Monopolar suction diathermy, coblation

Indications for SX(PYQ)

- Sleep apnea
- Chronic SOM
- Recurrent infections (sinusitis, AOM)
- Dental malocclusion

C/or Sx(PYQ)<3yr

- Velopharyngeal insufficiency
- Acute infection
- Bleeding diathesis

Corpplications of SX(PYQ)

- Hemorrhage (MC)
- Unmask Velopharyngeal insufficiency
- Grisel syndrome- Nontraumatic inflammatory atlantoaxial subluxation.

TONSILITIS

00:54:48

- Organism- Group A Beta hemolytic streptococcus
  - Has cross reactive antigens against myocardium, glomerulus & joints, resulting in complications like Rheumatic fever, Myocarditis, Subacute bacterial endocarditis, arthritis
  - Responsible for causing both acute & chronic tonsillitis

Types

Acute	• Duration of infection- <4 weeks • Acute superficial → Tonsils are red, with no enlargement • Acute follicular → Yellowish-white dots seen as infection spreads into the crypts

Acute	• Acute parenchymal → Tonsils are red, enlarged, and touch each other at the midline • Acute membranous → Dead bacteria and tonsillar tissue form a true membrane over the tonsils; no bleeding on peeling; membrane is limited to the tonsils
Chronic	• Duration of infection → > 12 weeks • Chronic follicular tonsillitis → Yellowish-white dots • Chronic parenchymal tonsillitis → Tonsils are red and enlarged • Chronic fibroid tonsillitis → Tonsils are small due to prolonged antigen-antibody reaction leading to fibrosis • Irwin Moore Sign ○ Used to differentiate chronic fibroid tonsillitis from normal tonsils ○ Pus oozes out on applying pressure on the crypts of the tonsils ○ Seen in chronic fibroid tonsillitis
Subacute	• Duration of infection- 4-12 weeks

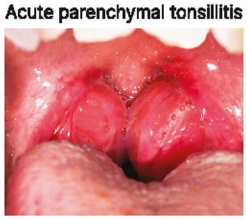
Acute superficial tonsillitis



Acute follicular tonsillitis



Acute parenchymal tonsillitis



Acute membranous tonsillitis



Symptoms

- Throat pain
- Fever- High grade, malaise, fatigue
- Dysphagia
- Odynophagia
- Drooling of saliva
- Difficulty in swallowing
- On examination- Cervical lymphadenopathy

Treatment

Medical treatment	• Acute tonsillitis
Surgical treatment (Tonsillectomy)	• Absolute indication of tonsillectomy is ○ Obstructive sleep apnea ○ Ca Tonsils • Relative indications of tonsillectomy ○ Chronic tonsillitis, recurrent tonsillitis (≥7 episodes in 1 yr, ≥5 episodes in 2 yrs or ≥3 episodes in 3 consecutive yrs) ○ 2nd episode of Quincy ○ Streptococcal infections • Non-tonsillar indications of tonsillectomy ○ Excision of the styloid process

	→ Elongation of the styloid process d/t calcification of the stylohyoid ligament → Eagle syndrome → External route or intra-oral route → Intra-oral route → Tonsil is removed to access the styloid ○ Palatal Sx → Uvulopalatopharyngoplasty (UPPP) → Reshape the uvula, palate, and pharynx in obese pts to treat sleep apnea ® Tonsil is removed as part of the Sx
Complications of tonsillitis	• Streptococcal-related complications • Neck abscesses (peritonsillar, parapharyngeal, retropharyngeal, and prevertebral space) • Referred otalgia
Complications of tonsillectomy	• Hemorrhage ○ Primary → Occurs intraoperatively ○ Reactionary → within 24 hrs of Sx → D/t slippage of the ligature → T/t- Emergency re-exploration and ligation ○ Secondary → After 24 hrs -14 days → D/t infections → T/t- IV antibiotics and watch, Re-exploration if bleeding continues • Injury to teeth • Injury to gums • Injury to TMJ • Aspiration (rare)

Membrane Over Tonsil

01:11:15

Tonsilitis	Diphtheria
• Pearly white in colour	• Greyish white in colour
• True membrane	• Pseudo membrane
• Avascular (from dead bacteria & tonsillar tissue)	• Vascular (from exotoxin liberated by the bacteria)
• No bleeding on peeling	• Bleeds on peeling
• Limited to tonsils	• Spreads
• No bull neck or toxemia	• Bull neck, Toxemia
	

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MCQs

01:13:30

Q. A child with a mild cold who also has adenoid facies has been identified as having chronic adenoid hypertrophy. Which is the best surgical management for this condition? FMGE July 2023

A. Coblator

B. Microdebrider

C. Curettage

D. Breathing exercise and symptomatic treatment

Ans. A.

Explanation:

- Coblation- Less tissue damage & less bleeding
- Microdebrider- Does not seal the cut ends
- Curettage- No control over tissue damage and bleeding
- Breathing exercise and symptomatic treatment- Recommended in mild enlargement

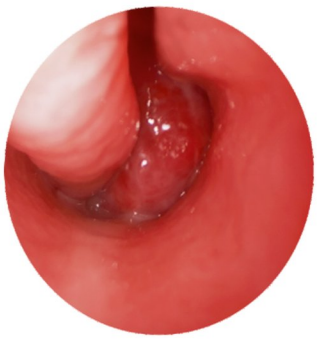
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YOURWISH

16. PHARYNX PART 2

JUVENILE NASOPHARYNGEAL ANGIOFIBROMA

00:00:01



- M/C benign tumor of the nasopharynx
- Benign but locally invasivae
- Most commonly seen in the juvenile age group
- JNA occurs only in males because the tumor is dependent on testosterone for its growth
- Site of origin
  - Sphenopalatine foramen → located 8-10mm behind the posterior end of the middle turbinate
  - Sphenopalatine foramen is related to the posterior wall of the maxillary sinus

Clinical Presentation

- Bleeding
- U/L SOM → CHL
- Frog Face → Proptosis, Widening of the nasal bridge
- Tumor spread superiorly → Sphenoid sinus → Cavernous sinus → nerve involved
  - Nerves palsy
    → IInd cranial nerve → At the roof of the sphenoid
    → IIIrd, IVth, Vth, VIth → Cavernous sinus
- Classification system
  - Radkowskis
    → Stage 1- Limited to the nose, nasopharynx, and sinuses
    → Stage 2- Orbital, pterygopalatine fossa, or infratemporal fossa involvement
    → Stage 3- Cranial involvement
  - FISCH
  - SESSIONS

Investigation

- Biopsy, FNAC, Probe test, Digital palpation → C/I
- HPE → JNA is a vascular tumor → Lacks Tunica media
- CECT → IOC
  - Holman Miller sign → Anterior bowing of the posterior maxillary sinus
  - Hondousa sign → widening of jap between the maxilla and mandible
- Axial CT
  - Chopstick sign → Erosion of the pterygoid base
- MRI → To know about the Intracranial extension
- Angiography → To identify the feeding vessel
  - M/C feeding artery → Internal Maxillary artery



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Treatment

- Angioembolization and surgical resection

NASOPHARYNGEAL CARCINOMA

00:06:20

- Bimodal age distribution → mostly affects the older population
- Males > Females
- Etiology
  - Chinese / Mongoloid population
  - Smoking, Tobacco
  - Smoke from an incense stick
  - EBV → Screening
    → IgA virus capsid antigen → most widely used
    → IgA against early antigen → most specific
- Site of origin
  - Fossa of Rosenmuller → lies behind the torus tubaris in the lateral wall of the nasopharynx

Clinical Presentation

- Neck nodes → cervical lymphadenopathy
- Blood-stained nasal discharge
- U/L SOM → CHL
- Nerve palsy → $II^{nd}, III^{rd}, IV^{th}, V^{th}, VI^{th}, IX^{th}, X^{th}, Xi^{th}$
  - Tumor invades the base of the skull and the skull foramina
- Horner's syndrome
- Trotter's triad
  - Conductive hearing loss due to SOM
  - V^{th} nerve palsy → Pain along the distribution of the Vth nerve
  - X^{th} nerve palsy → palatal paralysis
- HO'S triangle → site for lymphadenopathy
  - Medial end of clavicle
  - Lateral end of clavicle
  - Junction of neck and shoulder
- Classification
  - WHO
  - HO'S
- Variant
  - Well differentiated
  - Moderately differentiated
  - Anaplastic → poor prognosis

Investigation

- Biopsy → IOC

Treatment

- Early stage → Radiotherapy
- Late stage → Chemoradiation

INICET May 2025

Q. Which of the following is not seen in JNA?

a. Recurrent epistaxis
b. Unilateral nasal obstruction
c. Exclusively to adolescence boys
d. Biopsy under anaesthesia to diagnose

Ans: d

RANULA

00:18:34

- Extravasated cyst from the sublingual salivary gland
- Location → Paramedian
- It is due to injury to the duct of the sublingual salivary gland



Ranula
↓
Pierces the mylohyoid muscle
↓
Comes into the neck
↓
Plunging ranula

- Features of swelling
  - Translucent
  - Bluish and cystic swelling in the floor of the mouth
  - Transillumination test - Positive
- Investigation
  - History / Examination
  - USG
- Treatment
  - Marsupialization with excision of the sublingual gland

PHARYNGEAL SPACES

00:20:48

Peritonsillar Space

NEET PG 2024, INICET 2024, FMGE 2025

Tonsil is encapsulated → formed by pharyngobasillar fascia
↓
Superior constrictor muscle
↓
Buccopharyngeal fascia

- Space between Pharyngobasillar fascia and
- Superior constrictor muscle

Quinsy (Or) Peritonsillar Abscess

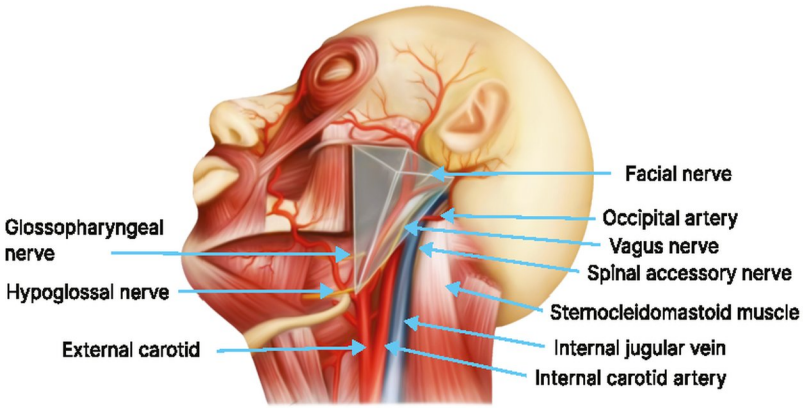
- Etiology- Tonsillitis
- M/C in the adult / elder population → because the crypts of tonsils are deeper and infection spreads into the peritonsillar space easily

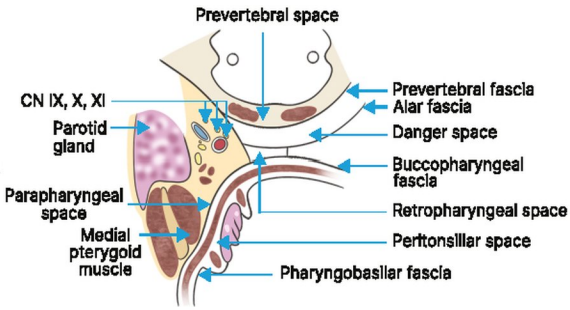
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- Clinical features
  - Throat pain
  - Difficulty in swallowing
  - Drooling of saliva
  - Hot potato voice or muffled voice
  - U/L bulge of tonsil with uvula pushed to the opposite side
- Treatment
  - Hot tonsillectomy
    → I&D + tonsillectomy
  - Interval tonsillectomy
    → I&D → wait for 4 to 6 weeks → tonsillectomy
    → Preferred

Parapharyngeal Space

- Largest space
- Boundaries
  - Superiorly → Base of skull
  - Inferiorly → Hyoid bone
  - Laterally
    → Angle of mandible
    → Masseter and Medial pterygoid
    → Parotid
  - Anteriorly and medially
    → Buccopharyngeal space
  - Posteriorly
    → Carotid sheath and its content
    – ICA
    – IJV
    – IXth, Xth, XIth, XIIth cranial nerve
  - Posteromedially communicates with the retropharyngeal space
- 2 compartments
  - Divided based on the styloid process
  - Pre-styloid compartment
  - Post-styloid compartment



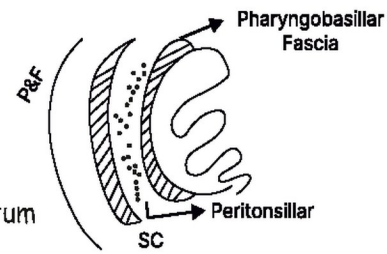


Pre Styloid Compartment Abscess

- U/L bulge of tonsil
- Neck swelling in the upper third of the sternocleidomastoid muscle

Post Styloid Compartment Abscess

- Bulge behind the posterior pillar



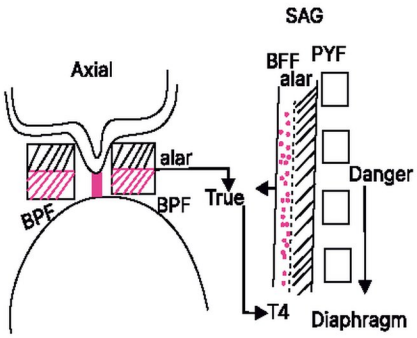
Retropharyngeal Space

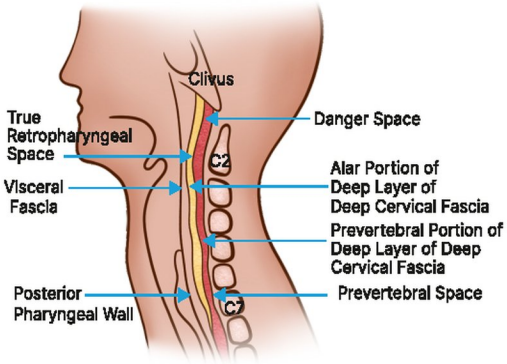
- Space behind the pharyngeal wall
- Buccopharyngeal fascia and Prevertebral fascia fuse in the midline to form a septum
  - Divides the retropharyngeal space into 2 compartments

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YOURWISH

- Boundaries
  - Anteriorly- Buccopharyngeal fascia
  - Posteriorly- Prevertebral fascia
  - Superiorly- Base of skull
  - Inferiorly- At the level of T4
- Divided into 2 spaces by the Alar fascia
  - Anterior compartment
    - → Space between buccopharyngeal fascia and alar fascia → True Retropharyngeal space → extends up to T4
    - → True Retropharyngeal fascia, AKA Space of Gillette. It contains the Node of Rouvier, which drains the tonsil and the adenoids
  - Posterior compartment
    - → Space between alar fascia and prevertebral fascia → Danger space → extend upto diaphragm → It can cause Mediastinitis





True Retropharyngeal Space Abscess

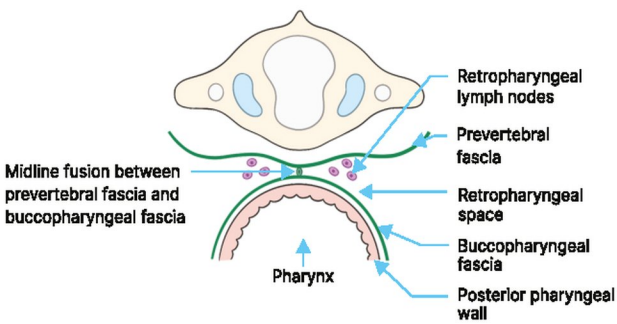
- Etiology
  - Infection from tonsil and adenoid
  - Common in children
  - Acute condition
- Clinical presentation
  - Present as a bulge on one side of the midline

Danger Space Abscess

- Etiology
  - TB Spine
  - Common in adult
  - Chronic condition
- Clinical presentation
  - Present as a bulge on one side of the midline

Prevertebral Space

- Boundaries
  - Anteriorly- Buccopharyngeal fascia
  - Posteriorly- Body of a vertebra
  - Superiorly- Base of skull
  - Inferiorly- Ends at the level of the coccyx
- Unpaired/single/Midline space



Prevertebral Abscess

- Etiology- TB spine
- Clinical presentation- Presents as a single swelling in the midline

Investigation Of Pharyngeal Space Infection

- CECT

Treatment Of Pharyngeal Space Infection

- Incision and Drainage
- IV antibiotics
- Secure the airway

One Liners

- Peritonsillar space is between the **pharyngobasilar fascia** and the **superior constrictor muscle**
- All the pharyngeal spaces extend superiorly from the **base os skull**
- Parapharyngeal space extends inferiorly upto **Hyoid**
- True retropharyngeal space extends inferiorly upto **T4**
- Danger space extends inferiorly upto **Diaphragm**
- Prevertebral space extends inferiorly upto **Coccyx**
- Smallest space is the **Peritonsillar**
- Commonest cause of a parapharyngeal abscess is **Tonsil**
- Node of Rouvier is **the retropharyngeal node**
- Space of Gillette is **the retropharyngeal fascia**
- Danger space is between the **Alar** fascia and the **prevertebral fascia**

Difference between Retropharyngeal abscess and Prevertebral abscess

	Retropharyngeal abscess		Prevertebral space
Extent	True compartment- Upto T4	Danger space- Upto diaphragm	Upto coccyx
Cause	True compartment- Tonsil / Adenoid	Danger space- TB spine	
Presentation	Bulge on one side of the midline		Single midline bulge
Diagnosis	CECT		CECT
Treatment	I&D, IV antibiotics, and secure the airway In case of TB → start ATT		I&D, IV antibiotics, and secure the airway

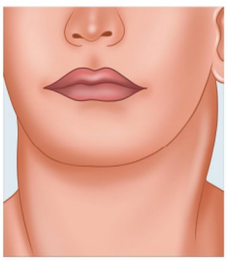
Peritonsillar Abscess

- U/L bulge of tonsil with uvula shifted to the opposite side



Parapharyngeal Abscess

- U/L bulge of tonsil with uvula shifted to the opposite side
- A/W with neck swelling

Retropharyngeal Abscess



- Swelling on one side of the pharyngeal wall
- Prevertebral soft tissue shadow is >2 times the body of the vertebrae
- Presence of air level/gases
- Reversal of normal cervical lordosis

Q. Which of the following is not true regarding acute retropharyngeal abscess?



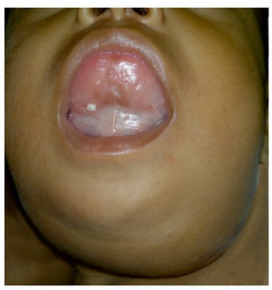
a. It is common in children
b. Produces difficulty in breathing and suckling
c. It is limited to one side of the midline
d. Incision and drainage are done from outside through the carotid sheath
e. It forms due to suppuration in the retropharyngeal lymph node

Ans: d

LUDWIG'S ANGINA

00:47:24

- AKA Vincent Angina
- Swelling in the floor of the mouth
- Pushes the tongue upwards and backward
- Goldring's criteria
  - It is Cellulitis, not an abscess
  - Always B/L in presentation
  - Presence of serosanguinous fluid and not pus
  - Spread is via tissue space, not by lymphatics
  - Sparing of the submandibular gland



- Source of infection
  - Infections from the tooth, gums, buccal mucosa, submandibular, and sublingual salivary glands
  - Mixed infection
  - Borrelia Vincenti
- Treatment
  - Presents with stridor
    → Tracheostomy
    → I&D
    → IV antibiotics
  - With no features of stridor

YOURWISH

→ Fibreoptic intubation
→ I&D
→ IV antibiotics

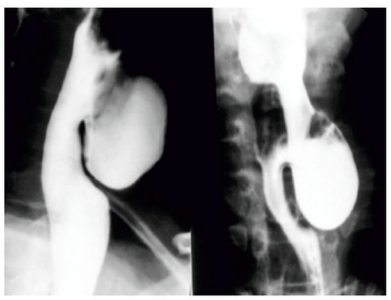
One Liner

- Etiology of Ludwig's angina: **Anaerobic / Mixed**
- Etiology of Vincent's angina **Borrelia Vincenti**
- **Woody hard** feel on palpation in Ludwig's angina
- **Mylohyoid muscle** divides the submandibular space into two compartments
- If the airway is endangered, treatment is **Tracheostomy**

PHARYNGEAL POUCH

00:51:31

- Outpouching of the mucosa through the Killian's Dehiscence
- Killan's Dehiscence
  - AKA Zenker diverticulum
  - Between Thyropharyngeus (Oblique fibres) and cricopharyngeus (Horizontal fibres) of Inferior Constrictor muscle
- False diverticulum
- Seen in the older population
- Etiology
  - Loss of satiety
  - Gurgling sound → BOYCE SIGN
  - Dysphagia
  - Sleep → Cough
- Investigations- Barium swallow
- Treatment
  - Dohlman's operation → Endoscopic diverticulectomy



ORAL AND OROPHARYNGEAL CANCERS

00:53:38

- M/C carcinoma → Squamous cell carcinoma
- M/C site worldwide → Lip (lower lip 98%)
- M/C carcinoma of upper lip → Basal cell carcinoma
- Prolonged exposure to sunlight is an important predisposing factor for carcinoma lip
- M/C site in India → Gingivobuccal sulcus
- M/C carcinoma of
  - Oropharynx → Tonsil
  - Hypopharynx → Piriform fossa
- Risk factor → smoking, tobacco chewing, repeated trauma by ill-fitting/sharp denture, alcohol, HPV
- Premalignant condition
  - Erythroplakia
  - Leucoplakia
  - Submucous fibrosis
  - Chronic hyperplastic candidiasis
- HPV (16, 18) → leading cause of carcinoma oropharynx, seen in the younger age group, and has a better prognosis than Non-HPV SCC

YOURWISH

YOURWISH

- Submucosal fibrosis pathology

Betel nut
↓
Presence of arecoline
↓
Normal tissue is replaced with fibrous tissue
↓
Blanching of the oral mucosa and trismus

- Submucosal fibrosis treatment
  - Intralesional steroids (Triamcinolone)
  - Hyaluronidase
  - Antioxidants
  - Cessation of betel nut chewing
- Lymph node metastasis in oral and oropharyngeal carcinoma is classified as Level I, II, III, and IV
- Midline tumors or tumors that cross the midline can have B/L lymph node metastasis
- M/C site of metastasis → Lung
- Verrucous carcinoma
  - A/K/A Ackerman's tumor
  - Well differentiated
  - Slow-growing SCC
  - Low malignant potential
  - Presents as → fungating / warty soft tissue papillary lesion of the oral cavity
- Plummer-Vinson syndrome
  - A/K/A Patterson-Kelly-Brown syndrome / Sideropenic dysphagia
  - Presents as a triad of
    → Post cricoid dysphagia
    → Hypopharyngeal/upper esophageal webs
    → IDA
- M/C carcinoma in females suffering from Plummer-Vinson syndrome is post-cricoid cancer

PYQ

00:57:10

Q. What could be the next step in managing the symptoms of a 7-year-old who underwent a tonsillectomy and now presents with snoring, palpitations, mouth breathing, slight hearing loss, high arched palate, and a decline in school performance, causing concern to the mother? FMGE June 2022

a. Adenoidectomy
b. Myringoplasty
c. Adenoidectomy with grommet insertion
d. Conservative management

Ans: c

Q. The image given below is from a chronic smoker's oral examination. Identify the lesion given. FMGE June 2021

a. Erythroplakia
b. Leucoplakia

c. Fordyce's granules
d. Koplik's spot



Ans: b

Q. What is the most common site of a ranula?

a. Floor of the mouth
b. Tonsil
c. Uvula
d. Cingula

Ans: a

Q. A 13-year-old boy presents with right-sided nasal obstruction and recurrent epistaxis for the past 6 months. What is the most likely diagnosis? INICET May 2024

a. JNA
b. Allergic rhinitis
c. Coagulation disorder
d. Antrochoanal polyp

Ans: a